IDE 402
DESIGN STUDIO IIX
PROJECT I

HOT BEVERAGE PREPARING
ELECTRONIC DEVICE

DESIGN MILESTONES

EMIR SERHAN BOZDAĞ

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INTRO

This document visually summarizes the key milestones of my design process for a hot beverage preparation device. It highlights how AI tools acted as clarifiers, re-framers, expanders, and critics throughout the project, and how I accepted, rejected, or transformed their suggestions as part of my own design decisions.

WEEK 1

Learning and assimilate AI as a design partner

WEEK 2

Research

WEEK 3

Research & Define

WEEK 4

Research & Define & Ideate

WEEK 5

Define & Ideate

WEEK 6

Prototype & Communicate

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PROCESS

RESEARCH
KEYWORDS

Daily rituals

Hot beverages

User habits

Pain points

DEFINE
KEYWORDS

Morning rush

Cleaning process

Counter clutter

Ritual quality

IDEATE
KEYWORDS

Alternative Workflows

Tactile interactions

Steam & Sound

Modularity

PROTOTYPE
KEYWORDS

Sequence testing

Feedback

Safety

Usability

COMMUNICATE
KEYWORDS

Scenerio

System diagram

User journey

Storyboard

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AI ROLES ACROSS THE PROCESS

PHASE	AI ROLE	SHORT DESCRIPTION
RESEARCH	CLARIFIER	Helped me cluster survey findings and highlight recurring user frustrations.
DEFINE	RE-FRAMER	Pushed me to describe the problem as a 'ritual experience' tension rather than only a functional issue.
IDEATE	EXPANDER	Generated alternative device forms, interaction flows, and multi-user scenarios.
PROTOTYPE	CRITIQUE	Pointed out inconsistencies in my system logic and missing feedback cues in the interaction.
COMMUNICATE	CLARIFIER/CRITIQUE	Supported me in simplifying the concept statement and sharpening the narrative of my final scenario.

Throughout the project, I used AI as a thinking partner rather than an automatic solution generator. It helped me see patterns, reframe my assumptions, and test ideas, but the final decisions about what to keep or change remained my responsibility.

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MILESTONE I UNDERSTANDING USERS & RITUALS

KEYWORDS

morning rush,
time pressure,
cleaning effort,
counter space,
sensory enjoyment

PROMPT (CLARIFIER)

Summarize the main problems and desires that appear in these survey answers about hot beverage preparation. Focus on time, cleaning, and the ritual feeling of the process.

AI RESPONSE SUMMARY

The AI grouped the issues into three clusters: time pressure in the morning, the difficulty and frequency of cleaning, and frustration with devices that occupy too much space for limited flexibility. It also highlighted that many users still value the sensory and ritual aspects of preparation, such as aroma, sound, and visible brewing. This showed me that the problem is not only speed, but the balance between efficiency and a meaningful ritual.

MY REFLECTION

Before this step, I was mainly focusing on speed and automation. After seeing the clustered insights, I realized that my concept should respect the ritual dimension instead of trying to remove all “slowness”. This shifted my attention towards designing a controllable tempo and richer sensory feedback, not just a faster machine.

DESIGN DECISION

- Accepted: “Morning time pressure and cleaning burden as primary design drivers.”
- Rejected: “Purely ‘one-button, fully automatic’ approach that eliminates user involvement.”
- Transformed: “Instead of only speeding up, I reframed the goal as ‘orchestrating a smoother and more intentional morning ritual’.”

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MILESTONE II

PROBLEM FRAMING & PRIMARY SCENERIO

KEYWORDS

single user at home, weekday mornings, limited counter space, shared kitchen clutter

PROMPT (RE-FRAMER)

Help me reframe this problem statement so that it emphasizes everyday ritual quality, not only technical performance. Use concepts like infra-ordinary, journey, and sensory experience.

AI RESPONSE SUMMARY

The AI encouraged me to describe the situation as an everyday ritual where the 'in-between' moments matter as much as the final drink. It suggested framing the device as something that choreographs body movement, sound, light, and aroma, instead of a hidden black box. It also stressed the importance of flexible rules so users can adapt the ritual to their own pace and style.

MY REFLECTION

This response helped me connect the project brief with the ritual design literature more concretely. I stopped thinking of my device only as a "machine" and started seeing it as a stage for small, repeatable actions that can be aesthetically and emotionally meaningful.

DESIGN DECISION

- Accepted: "Ritual-centered problem framing: balancing efficiency with expressive, sensory preparation."
- Rejected: "Generic, device-agnostic problem statements that could apply to any appliance."
- Transformed: "Turned vague 'better user experience' into a clearer focus on tempo, clutter, and sensory orchestration in one specific morning scenario."

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MILESTONE III

IDEATION & ALTERNATIVES

KEYWORDS

alternative flows,
multi-step interaction,
modular components,
shared use,
intuitive feedback

PROMPT (EXPANDER)

Based on this user profile and morning scenario, propose 3 different interaction logics and product architectures for a hot beverage device. Each should change the user's role and level of control in a different way

AI RESPONSE SUMMARY

The AI suggested three main directions: a highly guided, step-by-step ritual device; a modular system for parallel preparation of multiple drinks; and a more analog, hands-on brewer with enhanced sensory cues. It contrasted how each option affects user control, attention, and social interaction. This helped me see trade-offs between simplicity, expressiveness, and multi-user flexibility.

MY REFLECTION

I noticed that I was unconsciously favoring the most automatic option at first. After comparing the three AI-generated directions, I became more interested in a hybrid approach where the user still performs a few meaningful actions while the device coordinates timing and safety in the background.

DESIGN DECISION

- Accepted: "Hybrid interaction model: partial automation plus intentional user gestures."
- Rejected: "Fully manual ritual concept, because it did not address the time pressure of weekday mornings."
- Transformed: "Turned the 'modular multi-user' idea into a configurable but still compact system, suitable for a small shared kitchen.

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MILESTONE IV

CONCEPT SELECTION & SYSTEM LOGIC

KEYWORDS

main concept,
system diagram,
safety,
sequence,
feedback cues
(light, sound, steam)

PROMPT (CRITIQUE)

Here is my chosen concept and system diagram. Critically evaluate potential weak points in usability, safety, and feedback. Where might users be confused or frustrated?

AI RESPONSE SUMMARY

The AI pointed out possible ambiguities in the sequence of actions and in how the device communicates ‘ready’, ‘heating’, and ‘error’ states. It also questioned whether the cleaning steps were sufficiently integrated into the overall flow. This made me reconsider where to place visual and tactile cues in the interface.

MY REFLECTION

I realized that my first system diagram was very designer-centric and did not fully reflect the user’s perspective in a rushed morning situation. Integrating more explicit feedback moments and a clearer end-of-ritual signal became a priority.

DESIGN DECISION

- Accepted: “Need for explicit, multi-sensory feedback at key steps (start, brewing, ready, cleaning reminder).”
- Rejected: “Overly complex sequence with too many optional paths at the beginning of the interaction.”
- Transformed: “Simplified the early steps, but enriched the mid and end phases to keep the ritual feeling without overloading the user.”

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OVERALL REFLECTION ON AI IN MY DESIGN PROCESS

Throughout this project, AI functioned as a reflective surface for my ideas rather than a source of finished solutions. It helped me clarify user problems, reframe the brief as a ritual design challenge, and systematically compare alternative concepts.

At the same time, I experienced tool bias: certain tools pushed me toward highly polished, aesthetic concepts or overly complex systems. Keeping a structured log forced me to articulate why I was accepting or rejecting each suggestion, which protected my authorship and design intention.

In the end, my main contribution as a designer was to decide how much control, slowness, and sensory richness to keep in the ritual, given the real constraints of time, kitchen space, and user habits.

WHAT I ACCEPTED

- "Clustering of user pain points around time, cleaning, and clutter."
- "Ritual-focused framing that values sensory and embodied experience."

WHAT I REJECTED

- "Over-automated, 'one-button' solutions that ignore user agency."
- "Aesthetic-first concept directions that did not fit my scenario."

WHAT I TRANSFORMED

- "Modular and multi-user ideas into a more compact, scenario-specific system."
- "Generic AI suggestions into a concrete morning ritual for my defined user."

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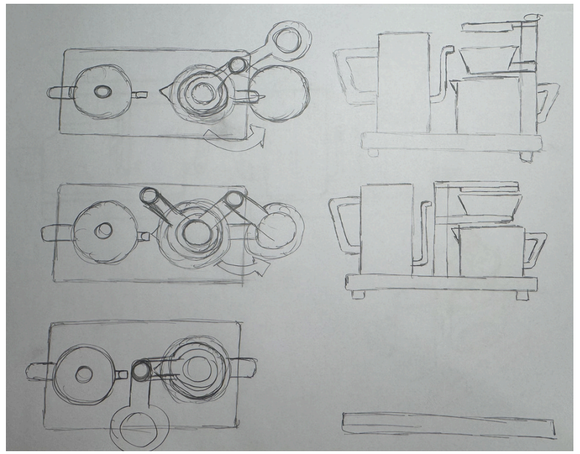
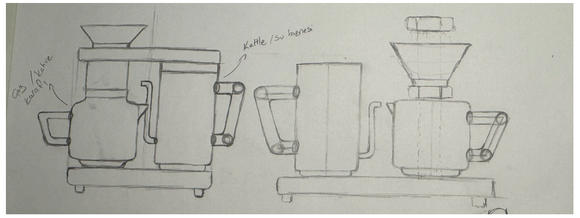
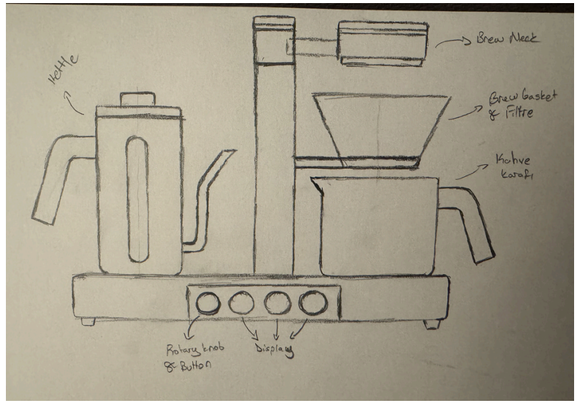
HAND SKETCHES & GENERATED VISUALS







These early sketches focus on how the device structures the preparation sequence: water filling, heating, brewing, and serving. I used them to test different ways of distributing actions between the user and the machine, especially under morning time pressure. They helped me visualize where to place key gestures and feedback cues.

I used AI-generated visuals to quickly explore different form languages and atmospheres around the device. Some images pushed me toward overly aesthetic or unrealistic concepts, but they helped me see how compactness, material contrast, and light could support the ritual feeling. I then simplified and grounded these ideas in my own sketches and system logic.